Quantum Cognition: Order Effects and Stress-Dependent Dephasing in a Quantum-Probability Evacuation Model

Muammal A. Fadhi, Mohamed Hassan, Bernard Kim, and Brodin Vance

***Abstract*—Human judgments can depend on cue order and uncertainty, but reproducing a judgment effect does not establish its consequences for action. Quantum Cognition embeds a two-qubit quantum-probability model in an evacuation simulator. Ambiguous cues rotate a density-matrix belief; direct danger observations reset it; stress controls dephasing. A deliberation unitary maps belief to protection. Calibration fixes a normalized overlap fraction; the resulting overlap coefficient changes from $\kappa = -0.439$ at the gambling conditionals to -0.785 in the volcano runs. At the gambling-calibration conditionals, reversing two cues in a specified window gives a signed protection gap of $+0.140$; the sign reverses (-0.190) under the simulator's initial priors. Maximal dephasing removes the specified calibration-window gap. Under deterministic action selection, 200 paired volcano seeds give a 0.55-step (66-minute) mean delay against a fully dephased control on 140 eligible pairs. A direct interference ablation gives a 0.689-step delay on 135 eligible pairs. Classical hesitation controls also produce delay. In that cue window, protection increases overall, but not monotonically for both orders, as dephasing reaches its maximum. Additional scenarios and archived sensitivity experiments show strong dependence on cue mapping and configuration. Qiskit circuit checks and simulated device noise support the implementation. These results characterize a modeling mechanism; they establish neither human behavioral validity nor a quantum computational advantage.**

***Index Terms*—Quantum cognition, order effects, disjunction effect, decoherence, phase damping, agent-based simulation, evacuation, Qiskit.**

I. Introduction

EVACUATION planning depends on models of how people interpret warnings and choose protective actions. People may seek further information before acting [1], and stress can affect decisions in ways that depend on the task [2]. Agent-based models represent such behavior through rules [3] or physical analogies, such as herding in a crowd [4]. These mechanisms do not by themselves specify how an agent's belief changes when evidence arrives in different orders.

Muammal A. Fadhi: Artificial Intelligence and Robotics Engineering, B.Sc. program, Al-Nahrain University, Baghdad, Iraq. Email: Muammal2006@gmail.com.

Mohamed Hassan: Electronics and Communication Engineering, Misr University for Science and Technology, 6th of October City, Egypt. Email: Mohamed.hassan505021@gmail.com.

Bernard Kim: Computer Science, University of Waterloo, Waterloo, Canada. Email: Bernardkim0805@gmail.com.

Brodin Vance: Honours Mathematics, University of Waterloo, Waterloo, Canada. Email: brodin.c.vance@gmail.com.

Written for Qiskit Fall Fest Waterloo, November 2026.

Code and archived results: https://github.com/Muammal/Qiskit-Fest-Qognition. Artifact versions and revision checks are identified in the Code and Data Availability statement.

Quantum probability offers a formalism for context-dependent judgment. Quantum cognition models reproduce question-order effects across national surveys [5] and disjunction effects, in which people reject a choice under uncertainty that they accept in either known condition [6], [7]. The framework has also been applied to inference [8], reinforcement learning [9], and open-system decision dynamics [10]. It is a mathematical account of judgment, not a claim that the brain is a quantum computer [11].

The question addressed here is narrower: what happens when a quantum-probability belief model is embedded in an agent whose decisions affect its simulated safety? The cited judgment models motivate the mechanism but do not establish its consequences in an evacuation. Recent quantum-like opinion models already include acting networks and non-commuting updates [12], [13]. Our focus is an emergency setting, a dephasing rate tied to the agent's stress, and a comparison with a fully dephased control in identically seeded worlds. Whether this stress mechanism describes human behavior remains an empirical question.

We address these questions with Quantum Cognition. Our contributions are:

1) **Order effects from non-commuting cues (C1).** For (storm, tremor, gas plume) minus (storm, gas plume, tremor), the protection gap is $+0.140$ at the gambling-calibration conditionals and -0.190 under the simulator's initial priors. Without dephasing, rotations about the same axis commute exactly.
2) **Stress as dephasing (C2).** For that window, the order effect decreases monotonically from 0.140 at zero dephasing to zero at maximal dephasing.
3) **An uncertainty dip (C3).** Calibration to a published disjunction effect gives an uncertain-case protect probability of 0.36, below the certain-case probabilities 0.69 and 0.59 and the classical prediction 0.64.
4) **Conditional hesitation in simulation (C4).** Across 200 volcano seeds under deterministic action selection, the quantum agent acts 0.55 steps (66 min) later than the fully dephased control on 140 eligible pairs (SE 0.091). The interference ablation isolates the readout term at matched inputs; classical hesitation controls show that delay is not unique to this mechanism. Equal observed survival is not an equivalence result.
5) **Robustness to simulated device noise (C5).** The order effect persists under a simulated IBM device noise model; all three reported circuit estimates lie within 2.01

standard errors of their exact values.

6) **Scenario dependence (C6).** Ten-seed Titanic, Pompeii and hypothetical chlorine-release runs show that coherence effects depend strongly on cue mapping. In Pompeii, one mapping gives approximately 1,063 deaths for the quantum mind and fewer than one for its control; another gives fewer than one for both.

We connect cue interpretation to an explicit quantum-probability representation and equivalent Qiskit circuits. The simulator exposes how that representation changes actions. We then use matched controls to distinguish the readout contribution from belief evolution and from delay mechanisms that also occur classically.

The remainder covers related work (Section II), the model and method (Section III), validation (Section IV), evaluation (Section V), discussion (Section VI), limitations (Section VII), and conclusions (Section VIII).

II. Related Work

A. Quantum Probability in Cognition

Quantum cognition represents a judgment as a state vector in a Hilbert space and a question as a projection [11]. Answering one question changes the state on which the next is evaluated, making judgments dependent on context and order without changing the underlying preferences. Busemeyer et al. [14] compared Markov and quantum models on the same decision data. Quantum-like Bayesian networks extend the approach to networks of beliefs [15], with belief entropy [16] used to quantify interference terms [17]. Yearsley and Busemeyer [18] provide a practical tutorial, and Pothos and Busemeyer [19] review the framework and its empirical limits. Dynamic decision models also predate Quantum Cognition: Busemeyer, Wang and Townsend [20] developed quantum and Markov random-walk accounts, while Kvam et al. [21] tested choice-induced changes in subsequent confidence during motion discrimination. Our fixed cue maps and evacuation setting are distinct from those evidence-accumulation tasks.

B. Order Effects and the QQ Equality

Wang et al. [5] analyzed 70 national surveys and two laboratory experiments in which reversing two questions changed responses. A quantum question-order model predicts a parameter-free constraint, the quantum question (QQ) equality, which the data supported. Wang and Busemeyer [22] derived and tested this prediction. Trueblood and Busemeyer [8] applied quantum probability to order effects in inference, where evidence arrives sequentially. Quantum Cognition is designed to exhibit this kind of order sensitivity. QQ agreement is nevertheless a limited diagnostic: Boyer-Kassem et al. [23], Section 3.4, distinguish it from the stronger Grand Reciprocity constraints of their non-degenerate model. Passing QQ alone does not establish that model's empirical adequacy, much less validate Quantum Cognition's cue maps.

C. The Disjunction Effect

In the two-stage gamble of Tversky and Shafir [6], 69% of participants chose to play again after a known win and 59% after a known loss, but only 36% when the outcome was unknown. This violates Savage's sure-thing principle. A related pattern occurs in the Prisoner's Dilemma [24]. Pothos and Busemeyer [7] accounted for these patterns through interference, while a matched Markov model could not reproduce them under its stated assumptions. Broekaert et al. [25] later compared heuristic, Markov and quantum-like accounts in a new two-stage-gamble experiment. Our interference parameter is calibrated to the original gamble.

D. Human Behavior in Emergencies and Under Stress

The Protective Action Decision Model [1] describes how people move from environmental and social cues, through threat perception, to protective action, including further information seeking. Conceptual models of human behavior in fire organize related findings for simulation [3]. Stress affects decisions, but its direction and magnitude depend on the task and moderating variables [2]. Classical pre-evacuation models already address cue interpretation and delayed action: Kuligowski [26] develops a protective-action framework, and Lovreglio et al. [27] model transitions between normal, investigating and evacuating states through random utility. These are relevant behavioral comparators; quantum probability is not required merely to represent pre-evacuation delay.

E. Open Systems and Learning

The general Markovian form of open-system dynamics is the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation [28], [29]. Our per-slot phase-damping channel is a discrete-time instance of dephasing dynamics [30]. Quantum-like decision models have used open-system dynamics to describe stabilization of a belief through interaction with its environment [10]. Li et al. [9] proposed quantum reinforcement-learning models of human decision-making.

F. Quantum-Like Agents

Quantum-like agents predate this work. Bisconti et al. [31] proposed a quantum belief–desire–intention architecture, and Fougères [32] described agents with superposition and entanglement properties. Yilmaz [33] used quantum cognition to simulate ethical dilemmas among agents with different perspectives. Bagarello et al. [10] modeled belief stabilization through interaction with a surrounding mental environment.

Recent opinion-dynamics models use density-matrix states, non-commuting survey operators and decaying coherence [12], or two-level states updated by non-commuting, partially dephasing channels whose order selects collective outcomes [13]. Qubit-encoded opinion dynamics have also been implemented on IBM hardware [34]. The closest of these share our central mechanism. Our proposed contribution is its use in simulated emergencies, with stress-dependent dephasing and a matched fully dephased control, rather than the quantum-like agent itself.

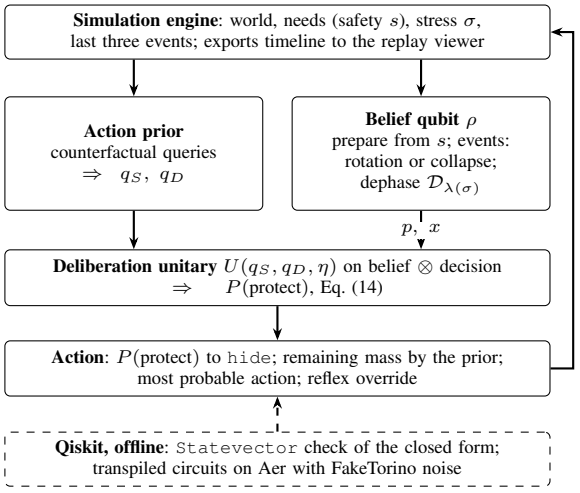


Fig. 1. Architecture. The engine supplies the agent's state. The belief qubit and action prior feed the deliberation unitary, whose output determines the protect probability. The chosen action returns to the engine. Qiskit verifies the closed form offline (dashed).

G. Agent-Based Simulation and Existing Tools

The social-force model [4] represents panic through a mixture of individual direction and herding, without an explicit belief about danger. Two open-source packages implement quantum cognition models: `quantum-mind` [35], including classical baselines and a QQ test, and `qcog` [36], including a decoherence parameter. The documentation consulted for this work did not embed these models in an emergency agent. Table I summarizes the comparisons.

III. System Model and Method

A. Architecture

Fig. 1 locates the model within the simulator. At each step, the engine passes the agent's needs, stress and three most recent events through a fixed interface. The mind returns an action, probabilities for all actions and a human-readable reason. The same interface serves the quantum mind and comparison minds (Section III-H). The simulation evaluates the decision model in closed form using NumPy [37]; Qiskit is used offline to check the closed form and execute circuits under simulated device noise (Sections IV and V).

B. Simulation Scenarios

The same mind is used, without modification, in four scenarios. Every scenario-specific cue must be mapped to one of the mind's event names. This modeling choice strongly affects results (Section VI). Constants without a cited source below are design assumptions.

Volcano. One agent occupies a synthetic 40×40 island of 250 m tiles over 36 two-hour steps. Unrest escalates through tremors and gas plumes to explosions, pyroclastic density currents, lahars, lava and ash fall. The sequence is motivated by the April 2021 La Soufrière eruption, St Vincent [38], but the island, timings and hazards are synthetic. Walking time follows Tobler's hiking function [39]. Hazards are precomputed for each seed and are not revealed to the mind. This scenario supplies the paired statistics in Section V-D.

Titanic. All 2,208 people aboard are simulated by class and crew department. Lifeboat launch times follow Wormstedt et al. [40]. Collision and tilt are mapped to tremors; stewards' warnings, crew orders and moving crowds to ambiguous gas plumes; and distress rockets, being told the ship will sink, and water at one's feet to collapses. A neutral-input setting removes two external nudges: a deck-dependent safety cap and a boost after choosing to hide. The historical comparison is 712 survivors of 2,208, or 32.2% to one decimal place [41].

Pompeii. A population of 11,500 experiences a chronology of the AD 79 eruption based on recent stratigraphy [42], [43]. At least 1,150 victims have been recovered at the site [44]; this is not the total death toll. Four mappings are compared: eruption column and pumice fall as eruption and ash-fall collapse (A); gas plume and ash-fall collapse (B, default); two gas plumes (C); or two tremors (D).

Waterloo. A hypothetical chlorine release affects 100 people at a community center in Waterloo, Ontario, laid out from OpenStreetMap outlines [45]. Exposure limits follow EPA Acute Exposure Guideline Levels [46] and the NIOSH Pocket Guide [47]; walking speeds follow evacuation observations [48]; and wind comes from modeled site weather [49]. Mappings treat the gas smell and alarm as gas plumes (A, default); the alarm as collapse (B); or the smell as tremor and alarm as gas plume (C). No real incident is implied.

C. Belief State

The model has a belief qubit, with $|0\rangle$ denoting safe and $|1\rangle$ danger, and a decision qubit introduced in Section III-G. The belief is a 2×2 density matrix rebuilt at each decision step. The mind is therefore stateless: history enters through the engine-maintained safety need $s \in [0, 1]$ (initially 0.85) and the three most recent events.

$$p = \mathrm{clip}(1 - s, 0.02, 0.98). \tag{1}$$

The belief is prepared by

$$|\psi_0\rangle = R_y(\theta_0)|0\rangle = \sqrt{1-p}|0\rangle + \sqrt{p}|1\rangle, \tag{2}$$

where $\sin^2(\theta_0/2) = p$. The clamp prevents the prior from reaching certainty; certainty arises from direct observation of danger (Section III-E). The density matrix is

$$\rho_0 = |\psi_0\rangle\langle\psi_0| = \begin{pmatrix} 1-p & \sqrt{p(1-p)} \\ \sqrt{p(1-p)} & p \end{pmatrix}. \tag{3}$$

This pure state has Bloch vector $(x, y, z) = (2\sqrt{p(1-p)}, 0, 1-2p)$. Initially, $p = 0.15$ and $(x, z) = (0.714, 0.700)$. A density matrix is needed because dephasing produces mixed states, which cannot be represented by a single state vector [30]. Danger probability is $\rho_{11} = \mathrm{tr}(P_D\rho) = (1-z)/2$; the off-diagonal terms encode coherence.

D. Event Gates

Each event in the three-slot window updates the belief. Ambiguous signs are modeled as rotations applied by conjugation:

$$\rho \mapsto U\rho U^\dagger, \qquad U \in \{R_x(\theta_e), R_y(\theta_e)\}, \tag{4}$$

TABLE I
PRIOR MODELS COMPARED WITH THIS WORK.

Model	Phenomena	Agent	Stress	Classical comparison	Runs on	Main limitation
Wang et al. [5]	Question order	No	No	Classical probability	—	Survey judgments only
Pothos & Busemeyer [7]	Disjunction effect	No	No	Markov model	—	One-shot choices
`quantum-mind` [35]	Order, interference	No	Not stated	Yes	Qiskit	No emergency agent
`qcog` [36]	Order effects	No	Decoherence param.	QQ check	Qiskit Aer	Not tied to stress
Helbing et al. [4]	Escape panic	Yes	Panic parameter	—	Classical	No belief state
Yilmaz [33]	Ethical dilemmas	Yes	Not stated	Not stated	—	No physical consequences
Chu [12]	Opinions, order effects	Yes (network)	Coherence decay	Classical limit	—	Opinions only
Niu et al. [13]	Opinions, order effects	Yes (network)	Fixed dephasing	None	—	Opinions only
This work	Order, disjunction, stress	Yes	Dephasing $\lambda(\sigma)$	Dephased control	NumPy; Aer	Hand-set constants

TABLE II
EVENT GATES (FROM `MIND/CONFIG.PY`).

Event	Gate	Angle (rad)
tremor, storm, animal	R_y	$\pi/3$
find food	R_y	$-\pi/4$
gas plume	R_x	$3\pi/5$
injury	R_x	$2\pi/3$
cold night	R_x	$\pi/6$
eruption, ash fall	collapse onto $\|1\rangle$	—
none	identity	—

where

$$R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \tag{5}$$

$$R_x(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}. \tag{6}$$

Unambiguous danger cues instead trigger collapse (Section III-E). Table II lists the event maps. R_y rotates the Bloch vector in the x–z plane, leaving y fixed; R_x rotates it in the y–z plane, leaving x fixed. Different-axis rotations generally do not commute [30]. Starting from $s = 0.85$ without dephasing, tremor followed by gas plume gives $P(\text{danger}) = 0.459$, while the reverse order gives 0.863. Same-axis rotations commute: tremor and storm give 0.984 in either order.

No listed angle equals $\pm\pi/2$. When a rotation acts on a diagonal belief, z becomes $z\cos\theta$; a $\pi/2$ rotation would therefore set $P(\text{danger}) = 0.5$ independently of the preceding diagonal belief.

These rotations are fixed geometric updates, not Bayesian updates. Their effect on danger probability depends on the current Bloch vector. A cue can therefore raise or lower alarm. In one run, a storm following three tremors lowered $P(\text{danger})$ from 0.850 to 0.366. These fixed rotations are a design departure from sequential question projections [5], [22]; they do not represent calibrated likelihood ratios. Repeated rotations are periodic: six uninterrupted zero-dephasing tremors return a persistent belief to its initial density matrix. The running mind instead rebuilds its belief from safety and a three-event window each step, so this identity is not a six-step prediction for the agent. Monotone accumulation of threat evidence is not guaranteed.

E. Collapse Events

Unambiguous danger cues are modeled by projection onto $|1\rangle$ [30]:

$$\rho \mapsto \frac{P_D \rho P_D}{\operatorname{tr}(P_D \rho P_D)}, \qquad P_D = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{7}$$

When the denominator is nonzero, this gives $\rho = |1\rangle\langle 1|$, danger probability one and zero coherence. A second collapse leaves the state unchanged. Thus, conditioning on this danger outcome erases the earlier belief. The event model uses this reset to represent direct observation of danger.

A large rotation can overshoot the danger state and rotate back toward safety; this reset cannot. Only eruption and ash fall are treated as unambiguous. Keeping strong but interpretable signs, such as injury, as rotations is a modeling choice, not a consequence of the simulated physics.

F. Stress as Phase Damping

Stress controls a dephasing channel rather than multiplying the final decision probability. The engine-maintained stress $\sigma \in [0, 1]$, initially 0.15 and increasing with threat signs and perceived peril, sets

$$\lambda = \operatorname{clip}\left(\frac{\sigma - 0.2}{0.7}, 0, 1\right). \tag{8}$$

Calm agents ($\sigma \le 0.2$) remain coherent, while agents with $\sigma \ge 0.9$ fully dephase. After every event slot, including empty slots, the belief passes through

$$\mathcal{D}_\lambda(\rho) = K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger, \tag{9}$$

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\gamma} \end{pmatrix}, \tag{10}$$

where $\gamma = 1 - (1 - \lambda)^2$. Diagonal terms are unchanged and off-diagonal terms are multiplied by $\sqrt{1-\gamma} = 1 - \lambda$. This parameterization makes λ a linear per-slot coherence loss and matches Qiskit Aer's `phase_damping_error` convention [50]. For events e_1, e_2, e_3, oldest first,

$$\rho_t = \mathcal{D}_\lambda\big(\mathcal{G}_{e_t}(\rho_{t-1})\big), \quad t = 1, 2, 3, \tag{11}$$

where $\mathcal{G}_e$ is the rotation or collapse map.

Dephasing removes coherence while preserving the safe–danger weighting. Depolarization would also shrink z, while amplitude damping toward $|0\rangle$ would make stress itself reassuring. This choice represents loss of coherence rather than loss of the diagonal belief.

At $\lambda = 1$, each slot ends with a diagonal state. Subsequent rotations then act through $z \mapsto z\cos\theta$; these scalar rotation updates commute. The first event still acts on a coherent prepared state, however. Reversing the first two events in a bare history can leave a residual order effect at maximal dephasing (Section VII). This commuting argument concerns rotation-only updates, not arbitrary reorderings involving collapse. For listed ambiguous rotations, $|z|$ contracts toward zero after the first slot, but a negative $\cos\theta$ can reverse its sign. Thus the state tends toward $P(\text{danger}) = 0.5$ with possible crossings; it need not approach danger certainty. For example, a gas-plume rotation takes a diagonal belief with danger probability 0.15 to 0.608. Collapse is a separate reset. These properties constrain interpretation of the cue maps.

G. From Belief to Action

A decision qubit has states $|C\rangle$ (carry on) and $|P\rangle$ (protect, or evacuate), and starts in $|C\rangle$. A real two-qubit unitary U models deliberation. Only $|u\rangle = U|S, C\rangle$ and $|w\rangle = U|D, C\rangle$ contribute, where S and D denote safe and danger.

Two classical conditionals are obtained from counterfactual queries to the action prior. For q_S, safety is set to 0.95 and threat events removed; for q_D, safety is 0.05 and an eruption is appended. Both are clamped to $[0.02, 0.98]$. The columns are

$$|u\rangle = \sqrt{1-q_S}|S, C\rangle + \sqrt{q_S}|S, P\rangle, \tag{12}$$

$$\begin{aligned}|w\rangle = \sqrt{1-q_D}\big(\cos\beta|S, C\rangle + \sin\beta|D, C\rangle\big) \\ + \sqrt{q_D}\big(\kappa|S, P\rangle + \sqrt{1-\kappa^2}|D, P\rangle\big).\end{aligned} \tag{13}$$

Orthogonality fixes

$$\cos\beta = -\frac{\kappa\sqrt{q_S q_D}}{\sqrt{(1-q_S)(1-q_D)}}.$$

Set $\kappa = \eta\kappa_{\max}$, where

$$\kappa_{\max} = \min\left(1, \sqrt{\frac{(1-q_S)(1-q_D)}{q_S q_D}}\right).$$

For $|\eta| \le 1$, this ensures $|\cos\beta| \le 1$. QR completion supplies the remaining columns with $U \in \mathrm{SO}(4)$, which admits a decomposition using at most two CZ-equivalent entangling gates [51].

For Bloch coordinates (x, y, z) and $p = (1-z)/2$, measuring protect gives

$$P(\text{protect}) = \underbrace{(1-p)q_S + pq_D}_{\text{classical}} + \underbrace{x\sqrt{q_S q_D}\,\kappa}_{k\ \text{(interference)}}. \tag{14}$$

The interference term vanishes when $x = 0$. Since U is real, y does not enter this readout.

The free interference parameter η is calibrated to Tversky and Shafir [6]: imposing $P(\text{protect}) = 0.36$ at $p = 0.5$, $x = 1$, $q_S = 0.69$, $q_D = 0.59$ gives $\eta = -0.7854$ and $\kappa_{\max} = 0.559$. At that calibration state, protection is less likely than in either certain condition. The transferred parameter is the fraction η of the allowed overlap, not a fixed protection decrement. Because $\kappa_{\max}$ depends on q_S, q_D, the calibration does not transfer the gambling effect unchanged. In the reproduced volcano batch, $\kappa_{\max} = 1$ and $\kappa = -0.7854$ at every quantum decision; at the gambling calibration, $\kappa \approx -0.4388$. The readout also depends on $x\sqrt{q_S q_D}$. This transfer is a modeling assumption.

Orthogonality and Cauchy–Schwarz constrain the overlap of the protect components, giving $|k| \le \sqrt{(1-q_S)(1-q_D)}$. For the Prisoner's Dilemma conditionals 0.97 and 0.84, with $p = 0.5$ and maximal allowed negative interference, the minimum is 0.836, above the observed 0.63 [7], [24]. If the belief weights are also varied, the minimum over all beliefs and allowed interference is 0.810. Thus, 0.63 remains unreachable; 0.836 is the equal-weight bound, not the global minimum.

We use *protect* for the binary readout throughout. Its engine action is `hide`: moving toward shelter or an assembly point in the volcano, attempting lifeboat boarding on Titanic, and leaving the hazard area in Pompeii and Waterloo. It is an action label, not a claim of actual evacuation completion. The protect group (`hide`) receives $P(\text{protect})$; the six carry-on actions share $1 - P(\text{protect})$ in proportion to the prior. The agent chooses the most probable action unless a reflex overrides it when thirst or warmth falls below a critical threshold.

H. Controls and Ablations

The primary *fully dephased control* is the same pipeline with $\lambda = 1$: safety input, event window, conditionals, action split, decision rule and reflexes are shared. Its code identifier is `classical_mixture`. It is a coherence control, not a fitted Bayesian model or a representative of all classical cognition.

Dephasing changes both belief evolution and readout interference. For (none, none, tremor) at $s = 0.85$, danger probability is 0.634 in the quantum mind and 0.325 in the control. The comparison therefore measures coherence throughout the pipeline, not k alone. The initial preparation remains coherent until the first slot ends. Once actions diverge, subsequent agent states and perceived cues can diverge.

An *interference-removed arm* keeps the quantum belief update and stress-dependent dephasing, but sets $k = 0$ only in Eq. (14). On identical input states, this changes only the readout; complete paired runs estimate its downstream effect, including feedback. A prior-only reference selects the largest prior action weight without a belief state.

A further classical hesitation control uses the dephased belief and

$$P_h = (1 - 4hp(1-p))\big[(1-p)q_S + pq_D\big].$$

The factor represents postponement under uncertainty and preserves both certain-case conditionals. We report $h = 0.4375$, chosen to match the same gambling calibration ($0.64(1 - h) = 0.36$), and $h = 0.2396$ fitted to the seven selected source-study rows in Table III [24], [25], [52]–[56]. These are phenomenological classical rivals; neither is a Bayesian evidence-accumulation model. The fitted value is the median of 70 clipped least-squares fits over all 3/4 and 4/3 split-halves of those rows. The dataset and fitting code are supplied. We retain the published compilation's rounded proportions; we did not reconstruct individual responses. Two selected rows have unknown-case proportions between the known cases, so these

are not seven independent demonstrations of a disjunction effect. No evacuation outcomes were used to fit h.

I. Alternative Deliberation Models

The code also re-implements six published approaches: (i) a continuous-time two-qubit Hamiltonian [7], [14]; (ii) a quantum-like Bayesian network whose interference is set by belief entropy [15]–[17]; (iii) GKSL open-system dynamics [28], [29], motivated by belief stabilization [10]; (iv) generalized Grover-reflection reinforcement learning [9]; (v) sequential Lüders projections for QQ-equality checks [5], [22]; and (vi) quantum Bayesian belief updating [8]. The first three can replace the default deliberation; outputs from (i) and (ii) are also logged for comparison. None supplies the reported scenario results. Internal checks appear in Section IV; comparative evaluation is future work. Source code credits the implementations followed.

J. Methodology Scope and Non-goals

Quantum probability is used as a modeling formalism [11], not as a physical account of the brain or a computational speedup. The simulation uses the NumPy closed form [37]. Qiskit [57] verifies equivalent circuit execution offline, and Aer uses a FakeTorino calibration snapshot [58] for simulated device noise. No job was submitted to IBM hardware. A Grover-coined discrete-time quantum walk [59], [60] and a separate unit-checked hazard framework exist in the repository but are not used in the reported simulations.

No constants are fitted to evacuation data. In the default quantum model, only η is calibrated to published gambling data [6]. Event angles, the stress mapping, counterfactual safety values and the needs-based action prior are hand-set. In the baseline and reproduced control batch, the action rule is deterministic `argmax`; variation comes from world seeds. The separately reported warning-time sweep samples actions and varies individual traits. Quantum and fully dephased runs use paired seeds, with a two-sided sign test for evacuation delay and McNemar's exact test [61] for survival. We report means, standard errors and eligible seed counts. Mean-delay intervals use $\bar{d} \pm 1.96\,\mathrm{SE}$ over paired seeds; they are descriptive intervals across the selected simulated worlds, not uncertainty bounds for human behavior. Exploratory direct-ablation comparisons are identified separately from the original C4 test.

IV. Validation

Acceptance criteria are encoded in the test suite. Exact identities—unitarity, calibration and agreement between implementations—use tolerances from 10^{-9} to 10^{-12}. Shot estimates must lie within four binomial standard errors, $\mathrm{SE} = \sqrt{P(1-P)/N}$, of the exact value. All 48 decision-model tests are reported as passing, including Aer and FakeTorino circuit tests. Our archived validation report records an environment of Qiskit 2.6.2, Aer 0.17.2 and IBM Runtime 0.51.0; we have not reconstructed that earlier installation. The earlier noise record names Qiskit 2.6.1, with Aer 0.18.2 and Runtime 0.51.0 documented in the archived setup. For the repeated closed-form experiments we used Python 3.12.10, NumPy 2.5.4, Qiskit 2.5.1 and Aer 0.17.2 installed; these repeated experiments did not rerun the noisy circuits or establish the older software-version claims.

Table IV lists nine passing checks, two unmet targets and three informational rows. The three noisy circuit estimates are within 2.01 SE of their exact values. This is a check under the stated simulated noise and acceptance criterion, not a hardware result. During idling, measured coherence tracks e^{-t/T_2} [30] multiplied by approximately 0.90–0.94, attributed to gate and readout errors in the calibration snapshot.

The Prisoner's Dilemma target is outside the model's allowed range under the stated conditionals (Section III-G). Other rows check re-implementations rather than the default model. QQ equality holds by construction for sequential binary projective measurements. The Huang belief-entropy result differs from its published value by about a factor of two; this discrepancy is unresolved. The external packages were not installed, so no independent package cross-check was performed.

V. Evaluation

Unless stated otherwise, results use the initial safety $s = 0.85$ and the Tversky–Shafir conditionals $q_S = 0.69$, $q_D = 0.59$. These analytic examples use the calibrated conditionals; their initial belief ($p = 0.15$) differs from the calibration belief ($p = 0.5, x = 1$). Full simulations instead obtain conditionals from their action prior.

A. Order Effects (C1)

Fig. 2 shows the Bloch paths for tremor (R_y) and gas plume (R_x) in opposite orders. Their different final beliefs produce different decisions. With a preceding storm, the window (storm, tremor, gas plume) gives $P(\mathrm{protect}) = 0.585$; (storm, gas plume, tremor) gives 0.445, a gap of 0.140. Without interleaved dephasing, same-axis rotations commute and do not generate this rotation-order effect.

B. Stress as Decoherence (C2)

Fig. 3 shows the Section V-A order effect as dephasing increases. The gap decreases from 0.140 at $\lambda = 0$ to 0.085 at 0.25, 0.043 at 0.5, 0.015 at 0.75 and zero at one. In this window, maximal dephasing removes sensitivity to the swapped pair. The pair follows a storm and therefore a dephased slot; the first-slot exception in Section III-F does not apply. Both protection probabilities end at 0.642, above their respective zero-dephasing values. This does not make protection monotonic: the tremor-first curve reaches 0.648 near $\lambda = 0.5$ before decreasing. The specified model predicts a reduced order gap and an overall increase at full dephasing, not a universal stress-induced delay.

C. Uncertainty Dip (C3)

Fig. 4 compares the pure-belief curve with classical total probability. The classical prediction interpolates linearly between 0.69 and 0.59. The calibrated quantum curve reaches

TABLE III

SELECTED ROWS USED TO FIT THE HESITATION CONTROL. a, b ARE THE TWO KNOWN-CONDITION PROPORTIONS; u IS THE UNKNOWN-CONDITION PROPORTION. PRISONER'S DILEMMA VALUES ARE TRANSCRIBED FROM [7], TABLE 2 (PDF P. 3); GAMBLE VALUES SUM ROUNDED PATTERN FREQUENCIES FROM [25], TABLE 2 AND EQS. (3)–(5) (PDF PP. 4–5). ORIGINAL STUDIES IDENTIFY PROVENANCE; THIS IS NOT A NEW PARTICIPANT-LEVEL REANALYSIS. THE ORIGINAL TVERSKY–SHAFIR GAMBLING CALIBRATION IS EXCLUDED FROM THESE SEVEN FITTED ROWS.

Study	Condition	a	b	u
Shafir and Tversky (1992) [24]	Prisoner's Dilemma	.97	.84	.63
Croson (1999) [52]	Mean of first two experiments	.67	.32	.30
Li and Taplin (2002) [53]	Prisoner's Dilemma	.83	.66	.60
Busemeyer, Matthew and Wang (2006) [54]	Prisoner's Dilemma	.91	.84	.66
Kühberger et al. (2001) [55]	Two-stage gamble, Experiment 3	.80	.38	.43
Lambdin and Burdsal (2007) [56]	Two-stage gamble, coin	.64	.47	.39
Broekaert et al. (2020) [25]	Experiment 1, $X = 2$	.45	.66	.60

TABLE IV

VALIDATION CHECKS. P = PASS, F = UNMET TARGET, I = INFORMATION (NO ACCEPTANCE CRITERION). NOISY ROWS USE SIMULATED FAKETORINO DEVICE NOISE, NOT IBM HARDWARE.

Check	Criterion	Result	Status
Calibration: $P(\text{protect})$ at $p = 0.5, x = 1$	0.36, within 10^{-9}	0.36	P
U orthogonal, $\det U = +1$	$\max \lvert U^\top U - I \rvert < 10^{-10}$	2.2×10^{-16}; $\det = 1$	P
Closed form vs. density-matrix evolution (200 random mixed beliefs)	within 10^{-12}	within 10^{-12}	P
Closed form vs. Qiskit Statevector circuit	within 10^{-9}	within 10^{-9}	P
Collapse yields $\lvert 1\rangle\langle 1\rvert$, idempotent	within 10^{-9}	within 10^{-12}	P
Same-axis events commute (tremor, storm; no dephasing)	equal p	0.984/0.984	P
Damping matches Aer `phase_damping_error`	within 10^{-9}	within 10^{-9}	P
FakeTorino, 3 circuits, 4,000 shots	within 4 SE	max 2.01 SE	P
Transpiled two-qubit gate count	$\leq$ 2 CZ	2 CZ, 11 SX, 10 RZ	P
Coherence decay vs. e^{-t/T_2} (FakeTorino qubit 10)	—	ratio 0.90–0.94	I
Prisoner's Dilemma disjunction effect	reach 0.63	0.836 bound at $p = 0.5$	F
QQ equality [5], [22], re-implementation	—	gap 1.1×10^{-16}	I
Huang QLBN belief entropy [17], re-implementation	published -0.942	-0.486	F
`quantum-mind`, `qcog` [35], [36] cross-check	—	not run (not installed)	I

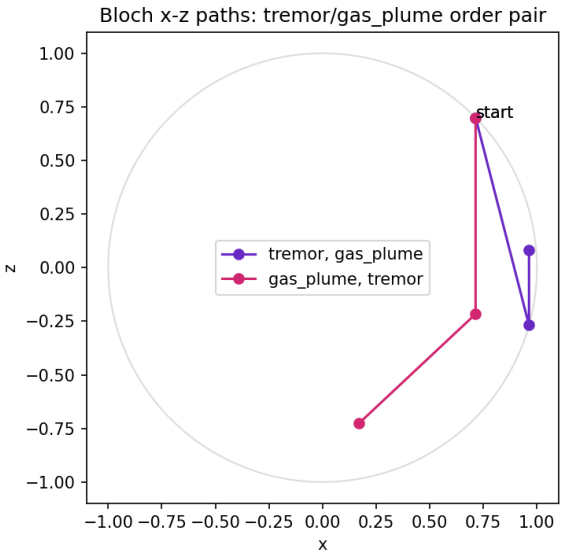


Fig. 2. Bloch x–z paths for a tremor and gas plume in both orders, starting from $s = 0.85$. This is the bare-pair illustration, without the storm prefix used for the 0.140 comparison in the text.

0.36 at $p = 0.5$, with a minimum of approximately 0.356 near $p = 0.59$. The value at $p = 0.5$ holds by construction because η was chosen to reproduce it. The remaining curve describes how this calibration extends to other pure beliefs; it is not an independent fit to human data.

D. In-Simulation Behavior (C4)

We reran seeds 0–199 at commit `ae818d8`, pairing quantum and fully dephased minds under `argmax`, and recovered the saved default-batch summary. Delay is the step index of the *first* protective action minus the index of the first volcanic sign, and is eligible only if both exist and the action is not earlier than the sign. A step is two hours.

On 140 eligible pairs, the quantum mind acts 0.550 steps later (SE 0.091; descriptive 95% interval 0.372–0.728 steps), equivalent to 66 min (44.6–87.4 min). It acts later on 45 seeds, earlier on five, and ties on 90. The exact two-sided sign test excludes ties, giving $p = 4.21 \times 10^{-9}$ on 50 non-tied pairs. Individual delays remain resolved only to two-hour steps; the fractional mean does not imply minute-level individual resolution.

The other 60 seeds are not unreported failures to act. Both minds first protect before the sign on 39 seeds; only the dephased mind does so on 20; only the quantum mind does so on one. Every mind has a sign and a first protective action. These seeds are outside the defined post-sign delay estimand; the conditional mean does not summarize timing for all 200 worlds.

Actions differ on 869 of 6,908 compared steps (12.6%), excluding 290 steps with a reflex override in either mind. Survival is 199/200 for each, with no discordant seeds (McNemar's exact $p = 1$). This near-ceiling outcome offers little

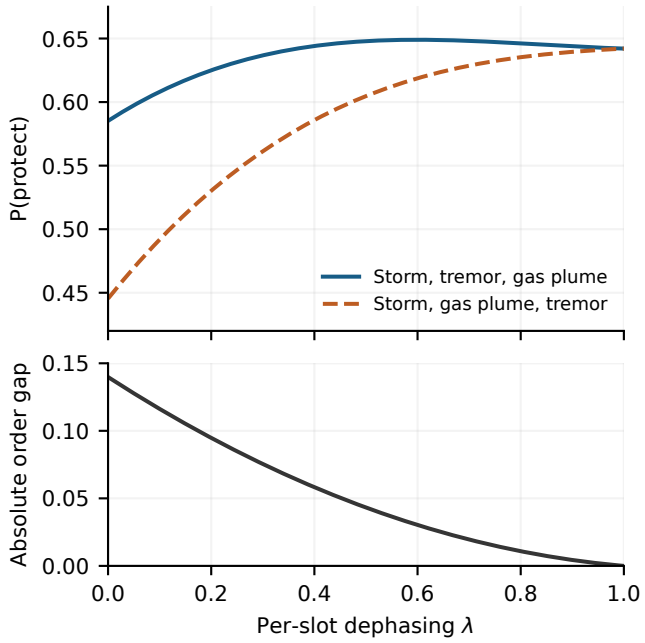


Fig. 3. Storm-prefixed comparison. Top: absolute protection probabilities for both cue orders. Bottom: their absolute gap. Parameters are $s = 0.85$, $q_S = 0.69$, $q_D = 0.59$ and calibrated η. The shrinking gap does not imply monotonic protection in both orders.

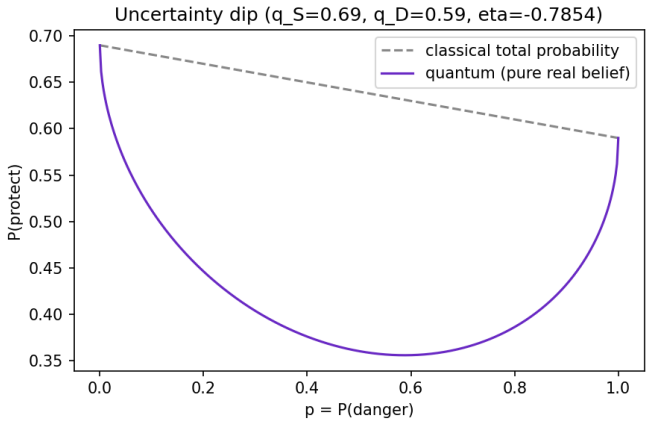


Fig. 4. $P(\text{protect})$ versus danger probability p for pure beliefs: the quantum model (solid) and classical total probability (dashed).

discrimination of survival costs and does not demonstrate equivalent performance.

Across those runs, stress-dependent dephasing is positive on 6,435 of 7,199 quantum decision steps (89.4%), spanning $\lambda = 0$ to 1. It therefore participates in the simulation, rather than appearing only in the analytic demonstration.

E. Simulated Device Noise (C5)

The decision circuits were transpiled for FakeTorino and executed with its noise model in Aer at 4,000 shots. The reference circuit uses two CZ gates. The archived experiment used Qiskit 2.6.1, Aer 0.18.2 and IBM Runtime 0.51.0, distinct from the later validation environment described in Section IV.

Unlike the storm-prefixed comparison in Section V-A, this experiment uses a tremor and gas plume without a preceding storm. Their two orders give simulated $P(\text{protect}) = 0.375$ and 0.567, compared with exact values 0.374 and 0.556. The exact gap is 0.182. All three circuit estimates, including the calibration case, lie within 2.01 SE of their exact values (Table IV). These are simulated device-noise results; no IBM hardware job was run.

F. Interference Ablation and Classical Hesitation

We evaluated all six arms in the same default batch (200 seeds per arm; Table VI). The direct quantum-versus-interference-removed comparison uses 135 jointly eligible seeds: quantum protects 0.689 steps later (SE 0.088; descriptive 95% interval 0.517–0.861), later on 53 seeds, earlier on none and tied on 82. Since only the readout term is removed at each matched input, the intervention distinguishes interference from coherent belief updating within this implementation. Its complete-run effect includes the trajectories that follow changed actions; it is not an instantaneous human causal effect.

The interference-removed arm acts 0.156 steps earlier than the dephased control on its 135 eligible pairs. Thus coherent belief updating without readout interference moves timing earlier in this configuration, while restoring interference delays it in the direct paired contrast. The Table VI means have different eligible cohorts and should not be subtracted. In an exploratory common-cohort calculation, all three arms are eligible on 135 seeds: interference-removed minus dephased is -0.156 steps, quantum minus interference-removed is $+0.689$, and quantum minus dephased is $+0.533$. Their seed-level differences add exactly on this common cohort. This makes the opposing directions explicit, including downstream trajectory feedback; it remains conditional on joint eligibility. Classical hesitation also produces delay: the $h = 0.4375$ arm gives 1.965 steps against the dephased control on 141 pairs. Thus hesitation is not unique evidence for quantum probability. A likelihood-based classical reasoner remains a distinct comparator.

G. Archived Configuration and Warning-Time Checks

The existing follow-up sweep uses sampled actions, 60 seeds, 30 replicates, eight warning leads (0–30 h), four starting distances and five arms: 288,000 recorded simulations. Per-person stress offsets and interference parameters vary across replicates. Saved analyses use seed-level bootstrap intervals and sign-flip tests, with Holm correction over eight leads within each control and metric. A rerun at the control batch's checkout and environment completed 8,011 of 9,600 cells before stopping. The full-sweep statistics reported here remain archived results; this is a separate experiment from C4. Their delay is to the first protective action at or after the first sign, even when protection also occurred before the sign. Runs with no post-sign protection are omitted from each arm's delay mean before seed-level contrasts. At zero lead, those fractions are 34.1% for quantum and 35.6% for the dephased control. This is a different conditional estimand from C4.

Quantum-minus-dephased mean-delay differences range from -0.084 to $+0.063$ steps and are not significant at any

TABLE V
SCENARIO OUTCOMES CALCULATED FROM SAVED PER-SEED RESULTS. MEANS AND SEED RANGES (PARENTHESES) WHERE AVAILABLE. VOLCANO USES 200 SEEDS; OTHER SCENARIOS USE TEN. MAPPINGS AND INPUTS ARE DEFINED IN SECTION III-B.

Scenario	Measure	Quantum	Dephased control	Prior-only	Reference
Volcano	Survival	99.5%	99.5%	100%	—
Titanic (default)	Survivors of 2,208	608.4 (570–675)	600.2 (554–664)	840.1 (768–888)	712 historical [41]
Titanic, neutral inputs	Survivors of 2,208	562.7 (537–597)	566.3 (515–629)	801.1 (773–856)	
Titanic, neutral, rockets ambiguous	Survivors of 2,208	296.1 (267–323)	270.2 (253–283)	801.1 (773–856)	
Pompeii, mapping A	Deaths of 11,500	0.1 (0–1)	0 (0–0)	0.1 (0–1)	
Pompeii, mapping B (default)	Deaths of 11,500	0.1 (0–1)	0 (0–0)	0.1 (0–1)	$\geq$1,150 recovered [44]
Pompeii, mapping C	Deaths of 11,500	1,063.2 (1,020–1,117)	0.7 (0–2)	0.1 (0–1)	
Pompeii, mapping D	Deaths of 11,500	0.2 (0–2)	0.1 (0–1)	0.1 (0–1)	
Waterloo, mapping A (default)	Time until half have left (s)	564	241	241	Hypothetical
Waterloo, mapping B	Time until half have left (s)	253	241	241	Hypothetical
Waterloo, mapping C	Time until half have left (s)	556	246	241	Hypothetical
Waterloo, mapping A, alarm off	People who left, of 100	24.5	91.0	—	Hypothetical

TABLE VI
REPRODUCED DEFAULT `ARGMAX` CONTROL BATCH, SEEDS 0–199. DELAY IS RELATIVE TO THE FULLY DEPHASED ARM, IN TWO-HOUR STEPS; n COUNTS ELIGIBLE PAIRS FOR EACH CONTRAST. L/E/T DENOTES LATER/EARLIER/TIED. THE DIRECT QUANTUM–INTERFERENCE-REMOVED CONTRAST IS REPORTED IN SECTION V-F AND USES ITS OWN JOINT COHORT.

Arm	Survival	n	Delay (steps)	SE	L/E/T
Quantum	99.5%	140	$+0.550$	0.091	45/5/90
Interference removed	99.0%	135	-0.156	0.062	6/24/105
Hesitation, $h = 0.2396$	98.5%	141	$+1.567$	0.126	95/0/46
Hesitation, $h = 0.4375$	98.5%	141	$+1.965$	0.131	111/0/30
Fully dephased	99.5%	141	$+0.000$	0.000	0/0/141
Prior only	100.0%	135	-0.919	0.115	0/57/78

lead (all adjusted $p = 1$). The preregistered quantum-versus-dephased survival-penalty claim at a lead of six hours or less was not supported. At zero lead, survival is 57.2% versus 54.9%, a $+2.3$ percentage-point difference (95% interval $+0.8$ to $+3.8$; adjusted $p = 0.048$); it is not evidence for the proposed survival penalty. This model result is not a general survival advantage. The action rule, traits, geometry and delay estimand change together, so the sweep is not a direct replication of C4 and does not isolate sampling alone.

The archived one-at-a-time analysis (seeds 0–99, `argmax`) also constrains C4. Scaling event angles by 1.2 gives a mean delay of 0.20 steps with descriptive 95% interval -0.045 to 0.445, including zero. The default for those 100 seeds is 0.479 steps, not the 200-seed mean. The direction of an order effect also depends on the conditionals: the storm-prefixed gap is $+0.140$ at the gambling values and -0.190 through the initial live-mind priors. These checks bound the delay claim to its declared configuration.

H. Scenario Mechanism and Alarm-Timing Sensitivity

We traced the person with the smallest ID in a prespecified seed-1, mapping-C Pompeii run with the full 11,500-person population. At each recorded quantum-run state, we also evaluated the dephased and interference-removed decisions without feeding them back into the world (Fig. 5). These same-input comparisons show how danger weighting and the readout term differ before trajectory feedback. A single local trace does not attribute the population death toll to one person's decisions. At the first non-reflex action difference, step 5 (75 min), the same recorded state gives protection probability 0.3667 without interference and 0.2304 with it. The readout term is therefore -0.1363: the quantum mind chooses `explore`, while both shadow controls choose `hide`.

We also varied Waterloo's staff-awareness-to-alarm delay over 0, 30, 120 and 300 s, retaining mapping A, the 100-person population, the leak and mobility assumptions, and the 60-s decision cadence. Staff-awareness time remains endogenous, so this parameter does not impose the same absolute alarm time across minds. Seeds 1–10 were paired across both minds and all delays (80 runs). All runs reached half off-site, and all 100 people left by the end. Table VII reports the half-off-site means and seed-paired differences. The default reproduces 564 versus 241 s. The gap remains positive for these four settings and grows to 470 s at a 300-s alarm delay; it is not a calibrated effect of alarm policy on real people.

VI. DISCUSSION

A. Design Rules

Three modeling lessons follow.

Rotation-order effects require non-commuting cue maps. Rotations about the same axis commute before dephasing is interleaved. Different-axis updates can generate order sensitivity, as in Section V-A. Interleaved channels and collapse maps must be considered separately; bare gate commutation is not a general statement about every complete event pipeline.

Cue mapping can matter more than coherence. Treating a cue as a rotation or a collapse changes how evidence is processed. In the neutral-input Titanic runs, changing distress rockets from collapse to an ambiguous cue reduces mean survivors from 562.7 to 296.1 for the quantum mind and

TABLE VII
WATERLOO MAPPING-A ALARM TIMING, TEN PAIRED SEEDS PER ROW. TIMES ARE SECONDS. INTERVALS ARE DESCRIPTIVE PAIRED-MEAN ±1.96 SE, NOT CONFIRMATORY TESTS. AEGL-2+ IS THE MEAN NUMBER IN THE MODEL'S TWO HIGHEST EXPOSURE CATEGORIES (QUANTUM/DEPHASED), NOT A PREDICTION OF CLINICAL OUTCOMES. MISSING HALF-OFF-SITE OUTCOMES: ZERO IN EACH ARM AND CONDITION.

Alarm delay	Quantum half-off	Dephased half-off	Paired gap	95% interval	AEGL-2+
0	552	223	329	286.2–371.8	0.6/0.0
30	564	241	323	283.3–362.7	0.5/0.0
120	630	294	336	299.7–372.3	0.6/0.0
300	786	316	470	419.9–520.1	0.9/0.0

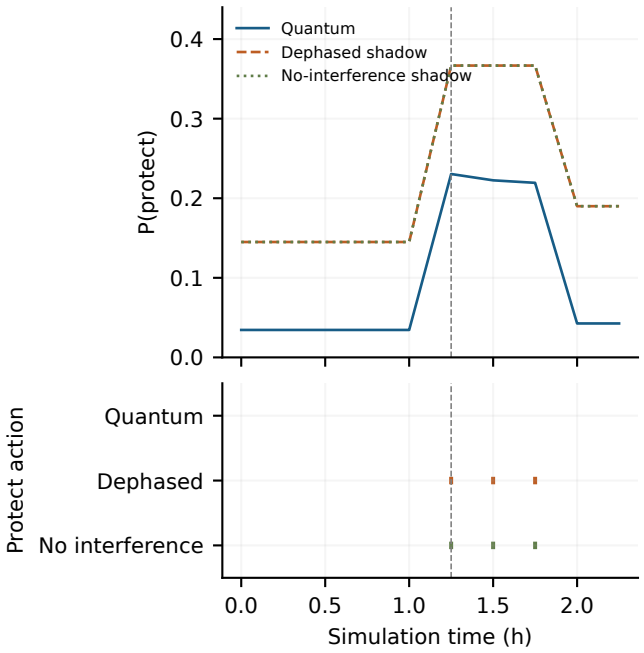


Fig. 5. First ten prescribed decision rows for person 0 in the seed-1 Pompeii mapping-C run; all 79 rows are retained in the supplement. Shadow controls evaluate the same recorded state. The lower panel marks protective actions; controls do not feed back. This is a local mechanism trace, not a population-effect estimate.

from 566.3 to 270.2 for its control, while the prior-only result stays at 801.1 (Table V). Collapse affects both belief-based minds, so this is not a coherence-only effect. The Titanic is a calibration scenario, not an independent validation event.

Dephasing removes interference at readout. At $\lambda = 1$, coherence is removed after every slot, so the readout interference term vanishes. The storm-prefixed rotation-order gap also vanishes (Section V-B), subject to the first-slot exception. Interpreting this as a consequence of human stress is a testable modeling proposal. The empirical effects of stress are task-dependent [2].

B. Results Across Scenarios (C6)

Table V combines the 200-seed volcano experiment with ten-seed runs for the other scenarios. In the default volcano batch, the observed delay accompanies no survival difference; the ceiling limits that comparison. In the default Titanic mapping, the quantum and fully dephased means differ by approximately eight survivors, well within their seed ranges. Both are below the historical 712; the prior-only mind, which chooses to hide more often, has the highest survival.

The Pompeii coherence difference depends on mapping. With both cues mapped to gas plumes (C), the quantum mind loses 1,063.2 people on average versus 0.7 for the control. With both mapped to tremors (D), both means are below one death. At the hypothetical Waterloo site, an ambiguous alarm (A) gives 564 s (9.40 min) until half the quantum agents have left, compared with 241 s (4.02 min) for the control. A collapse alarm (B) reduces the quantum time to 253 s (4.22 min). Without the alarm, 24.5 of 100 quantum agents leave, compared with 91.0 control agents.

These results characterize the model, not historical behavior. Default Pompeii and Waterloo mappings were chosen after inspecting results; all tested mappings are therefore reported. The Titanic assumptions have not been tested individually in full. The recovered Pompeii victim count is not a death toll and is not a validation target for the simulated population.

C. Generalization

The mechanism is portable across scenarios: non-commuting updates can generate order effects, and a negative interference term can lower protection for coherent beliefs. Dephasing suppresses the interference, with order effects depending on the event history and channel placement. The sizes and practical consequences depend on the cue maps, action rule, priors and warning scenario. The practical implication is to test how alarm interpretation changes a model's output before interpreting differences between cognitive mechanisms as behavioral predictions. Choosing these maps from data is necessary before making quantitative behavioral predictions.

VII. LIMITATIONS

Hand-set parameters. Event angles, the stress map and counterfactual safety values are design choices. The calibrated η is transferred from gambling to evacuation as a fraction of allowable interference. The archived angle sensitivity and classical hesitation arms show that a positive delay is not a robust or distinctive signature across all configurations.

Hand-specified action prior. The weighting of needs and events is not fitted to time-use or evacuation data.

Binary belief and decision. The model distinguishes safe from danger and protect from carry on. Richer beliefs and decisions need an extended representation.

Cue mapping. Mapping an observation to rotation, axis or collapse is hand-set and can determine outcomes. Collapse also erases the earlier belief, simplifying human certainty.

First-slot exception. The initial belief is coherent even for the fully dephased control. Reversing the first two events in a bare rotation history can therefore leave order sensitivity at maximal stress. The zero-order-effect claim applies to specified windows with a preceding dephased slot, not all histories.

Coverage of human data. The default model cannot reach the cited Prisoner's Dilemma target under the stated conditionals. One comparison re-implementation also has an unresolved discrepancy with its published value.

Simulated hardware only. Noisy results use a calibration snapshot within Aer. The exported circuits have not run on quantum hardware.

Scenario assumptions. Paired statistical tests cover only the volcano. The other scenarios use ten seeds and many unsourced assumptions. Changing walking speed on pumice, for example, moves Pompeii deaths under mapping C from about 1,050 to over 11,000. One-at-a-time variations do not establish combined sensitivity. The Waterloo sensitivity varies only the alarm delay after staff awareness; it does not jointly vary detection, hearing, dispersion and mobility. The minute conversions do not change its decision-time resolution.

No human-subject validation. Comparisons use published studies, not new participants. A historical shipwreck benchmark [41] is preregistered in the repository but has not been run.

Rule-based social cues. Pompeii and Waterloo transmit household or local word-of-mouth information and movement-derived crowd cues between agents. These use fixed rules. The decision module represents each person's belief separately and does not learn social-influence parameters or represent coupled cognitive states.

Shared mechanism. C2 and C3 both depend on coherence, so they are not independent lines of empirical evidence.

VIII. Conclusion

Quantum Cognition makes a quantum-probability decision mechanism inspectable through a simulator and equivalent Qiskit circuits. A specified cue reversal produces a $+0.140$ protection-probability gap at the gambling-calibration conditionals, while its sign reverses under the initial simulator priors (-0.190). Dephasing removes the specified calibration-window gap. The 0.36 uncertainty dip is a calibration result. The reproduced default volcano shows a 66-minute conditional mean delay against a dephased control and a direct 0.689-step delay against interference removal, without a demonstrated survival cost.

Classical hesitation controls also delay, while archived changes to angles and scenario configuration limit generalization of the positive default delay. Cue translation can alter outcomes more than the choice between coherent and dephased updates. For warning-model design, the useful result is therefore a transparent way to expose and test assumptions about ambiguous cues, rather than an established prescription for human evacuation. Human data, independently selected cue mappings and calibrated social-influence mechanisms are needed before quantitative behavioral claims. Hardware execution and the preregistered shipwreck benchmark remain future work.

Code and Data Availability

The public repository is https://github.com/Muammal/Qiskit-Fest-Qognition. Original reported summaries are available in the main-branch snapshot `0e68d5b`. The revision's control reproduction uses checkout `ae818d8`; the accompanying supplement contains source hashes, environment versions, the runner, seed-level records and descriptive contrasts. The supplement distinguishes archived inputs from the retained partial warning-time rerun and the new scenario checks, with protocols and environment records. A companion notebook uses a bare-pair order demonstration and a different seed cohort; its displayed gaps are not the storm-prefixed C1 experiment. No new human-participant dataset was collected.

Funding and Competing Interests

No funding was received for this work. The authors declare no competing interests.

AI-Assisted Revision

AI tools assisted language editing, literature discovery and computational checks. The authors remain responsible for the manuscript, its sources and its claims.

Author Contributions

The authors jointly contributed to conceptualization, project development, experimental evaluation and manuscript preparation. All authors reviewed and approved the manuscript.

Acknowledgment

We thank the organisers and mentors of Qiskit Fall Fest Waterloo. We also thank the authors of the published models and public implementations that our code re-implements (Section III-I). Map data are © OpenStreetMap contributors, available under the Open Database License. Weather data are by Open-Meteo.com [49].

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